9 November 2026

Dear Editors of Information Sciences,

Please consider our manuscript, “CPOR: Cumulative Protection for Online Reconciliation with Nodewise Error Limits,” as an original research article.

Forecast reconciliation can improve aggregate accuracy while increasing errors in individual regions. CPOR addresses settings in which forecasts must satisfy aggregation constraints and each observed node has a cumulative error limit relative to an agreed causal reference. For nonnegative outcomes without a known upper bound, we derive the exact one-step admissible set. Updating the remaining allowances using errors of the forecasts actually released maintains the limits at every prefix. Statistical and sequential forecast combinations provide two implementations of the framework.

The study combines a necessary-and-sufficient geometric characterization with experiments on seven datasets and three forecasting backbones, each trained three times. On identical reconciled forecasts from PIR, full projection reduces mean-node RMSE relative to common shrinkage by 3.57% in window tests and 4.34% in continuous tests. The corresponding gains over two-group shrinkage are 2.72% and 3.28%. We report protection costs, regional and system-level deterioration, and cases where unprotected PIR is more accurate. These paired comparisons distinguish the protection operator’s contribution from differences between complete forecasting procedures.

The paper fits the journal’s coverage of intelligent information processing, statistical techniques, algorithms and decision support, and its balance of theory and practice. The submission includes the manuscript, supplementary results, highlights, editable sources and a compact numerical evidence package.

Thank you for considering the manuscript.

Yours sincerely,
Lei Zhang
Capital Normal University, Beijing, China
lei.zhang@cnu.edu.cn